

Primitive Data Types



COMIXPLAIN

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Project Team:

Victor-Adriel De-Jesus-Oliveira
Hsiang-Yun Wu
Christina Stoiber
Magdalena Boucher
Alena Boucher

Contact:

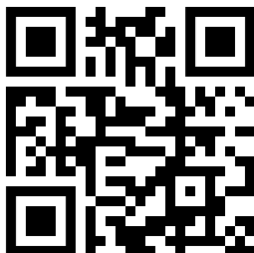
victor.oliveira@ustp.at

Illustrations:

Magdalena Boucher & Alena Boucher



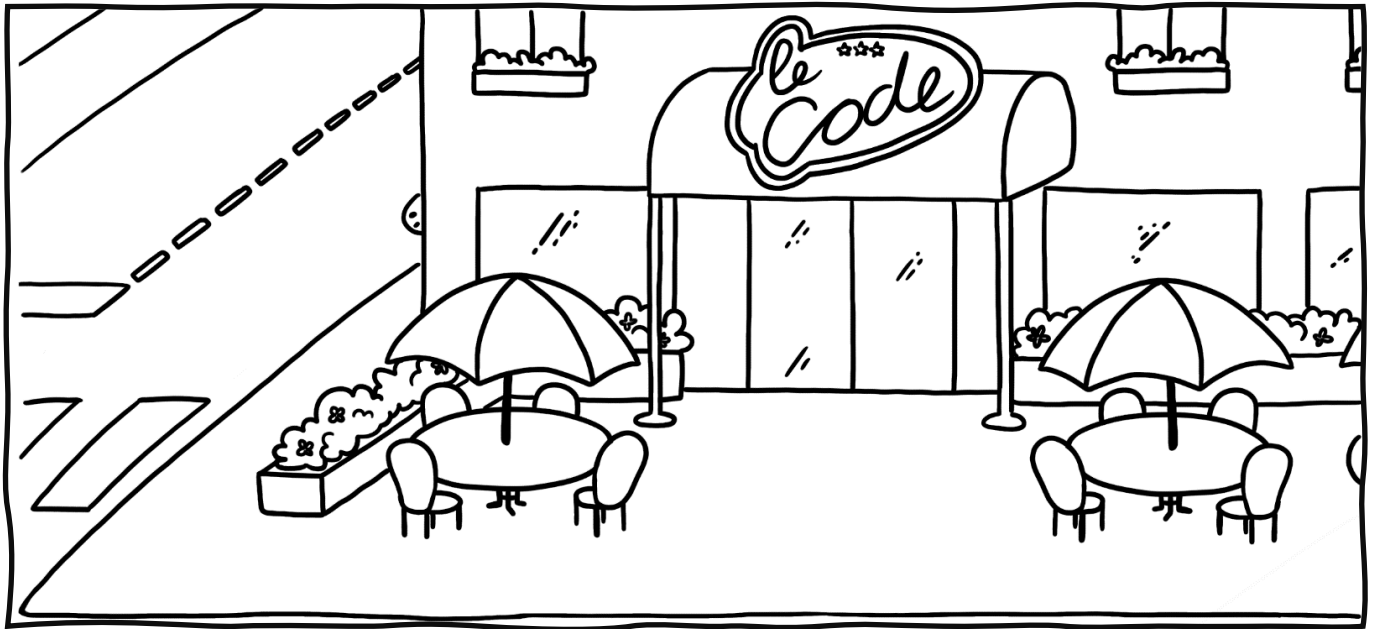
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Oh, you must be the new hire our manager was talking about!



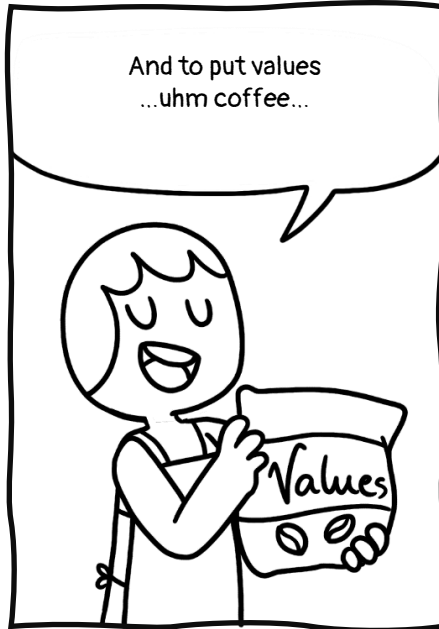
What do you mean you're not? Of course you are! Here's your apron. Come with me and I will show you the ropes quickly.



Your job is to execute orders from the program, uh, I mean the kitchen...

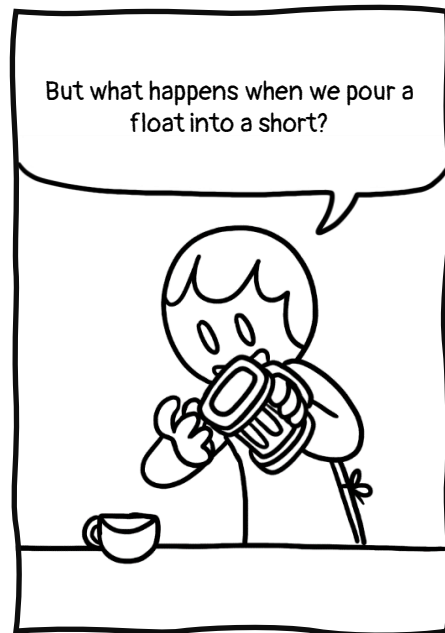
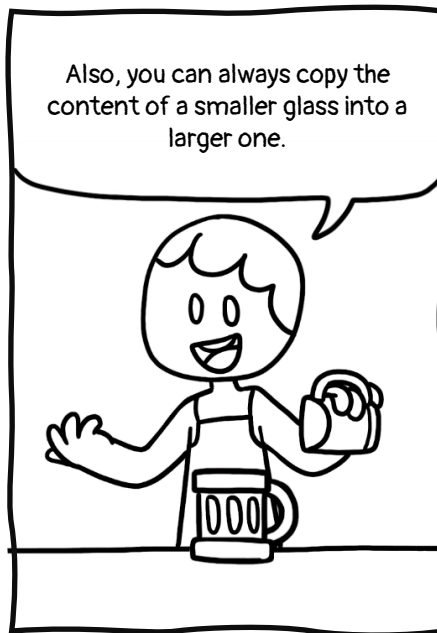
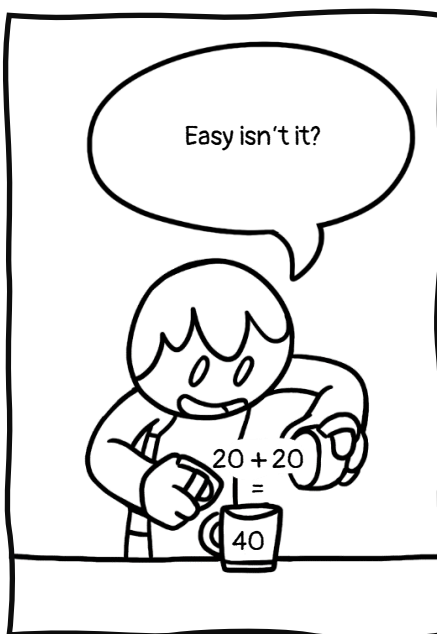
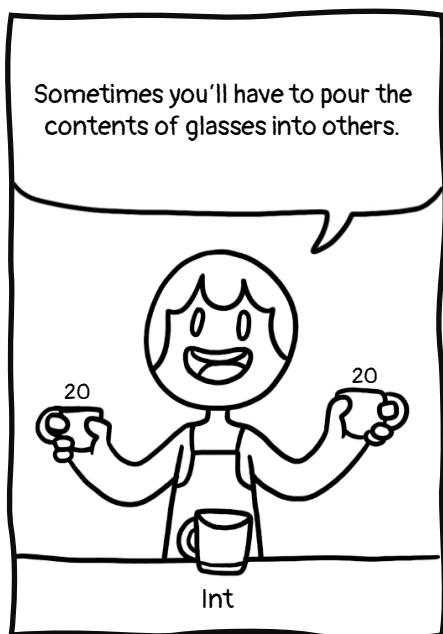
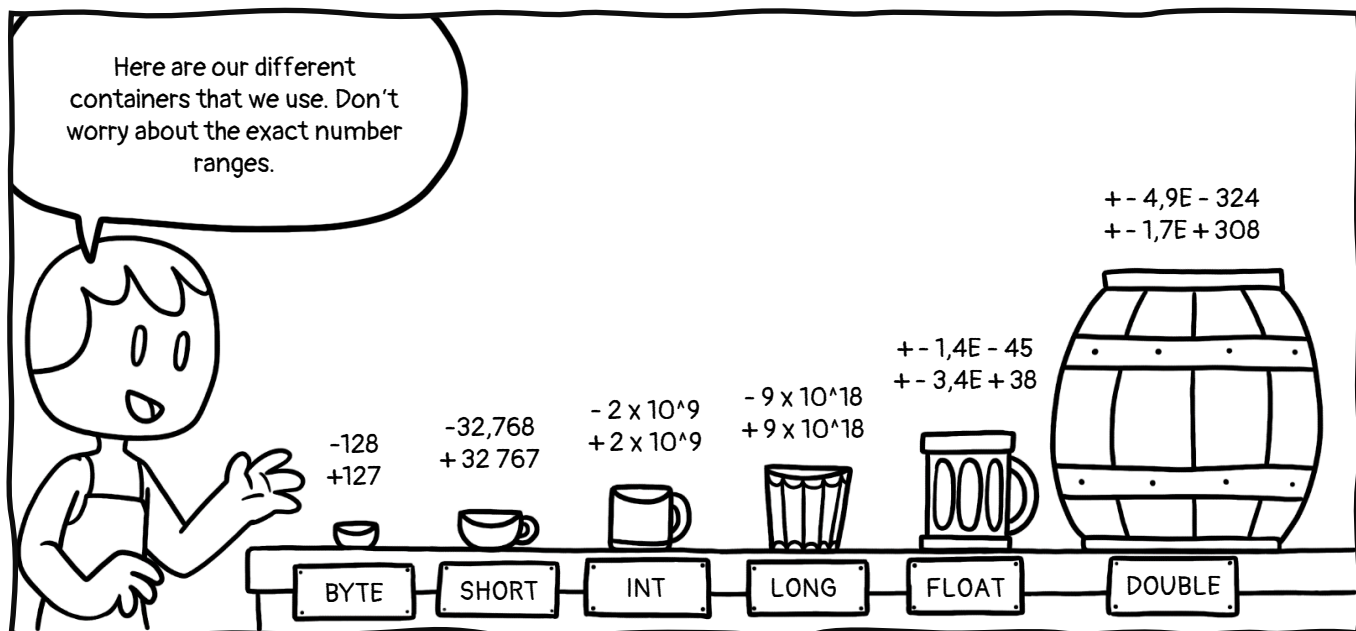


And to put values ...uhm coffee...



Into containers!





Only a part of it will fit.

And the rest will leave a huge mess on the counter.

And whether that's bad depends on the manager you're working with,

See this person here? She won't be bothered much and will clean it up for you.

But NEVER do this under his watch. He will throw a fit!

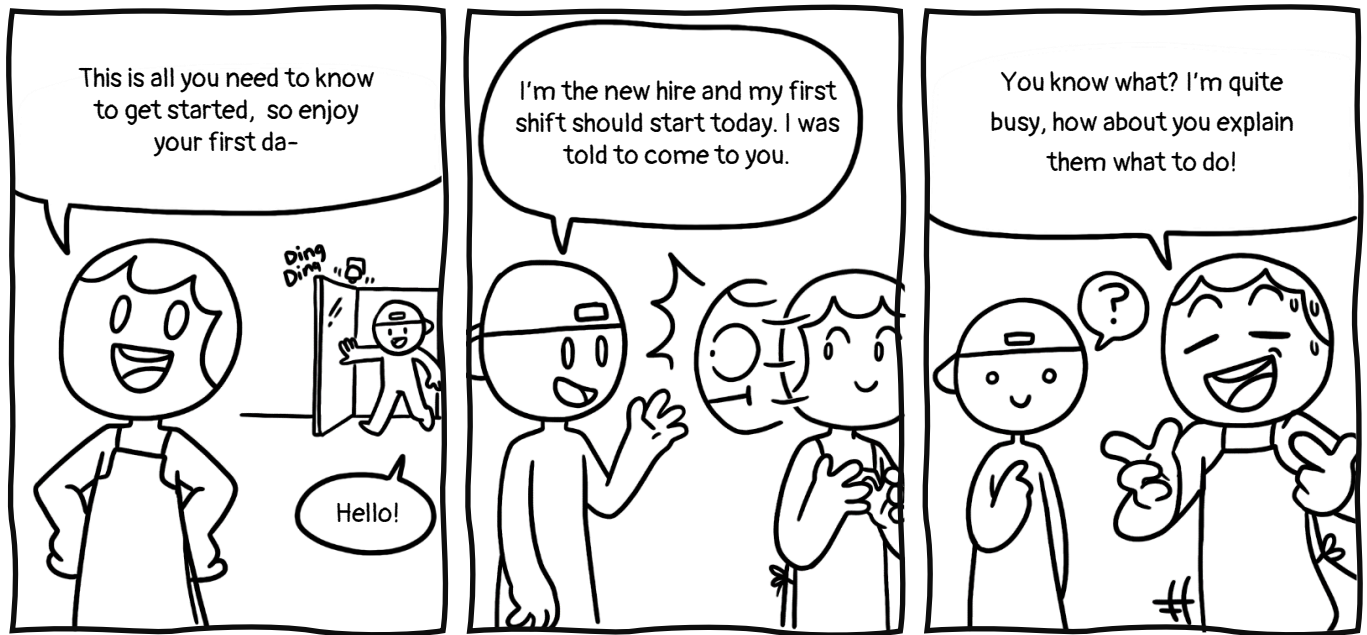
So depending on what you are working on, try to avoid it.

You may ask yourself why don't we just put everything into doubles.

...but there wouldn't be much space left on the tables...

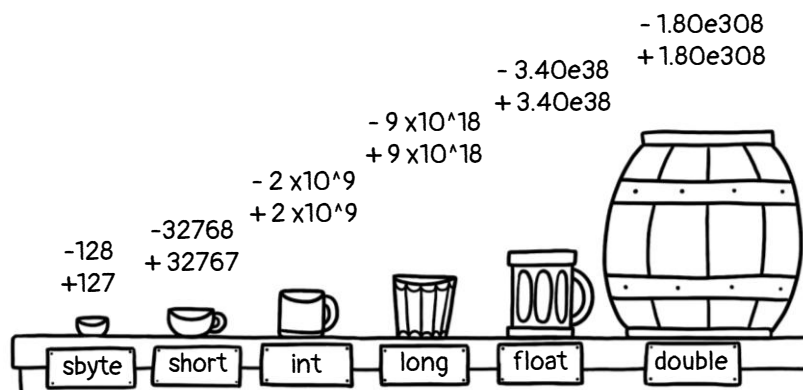
So please choose appropriate sizes.

Primitive Data Types



How would you explain the concept of primitive data types to the new employee? Use this space to make notes and drawings.

C# (C-Sharp)



C# also includes:
bool: a boolean value (true or false).
char: a Unicode character (U+0000 to U+FFFF).

Primitive types are predefined by the programming language itself. Here are the equivalent types and their ranges for other languages.



Javascript

2 1.3 Number: Represents all numeric values*



Boolean: Represents a boolean value (true or false)



String: Represents a sequence of characters



Undefined: Represents an uninitialized or undefined value



Null: Represents an intentional absence of any object value

*The type "number" includes integers and floating-point numbers. JavaScript doesn't have specific types like byte, short, or long.

Be aware that primitive types provide the programmer with essential data types for storing and manipulating data efficiently, often optimized for performance and memory usage.



Python

2 int: Represents arbitrary precision signed integers.

1.3 float: Represents floating-point numbers*



bool: Represents a boolean value (true or false).



str: Represents a sequence of Unicode characters.



NoneType: Represents the absence of a value, often used as a placeholder or default return value.

*The range of float is approximately 1.7×10^{-308} to 1.7×10^{308}

While many languages share similar primitive types such as integers, boolean values, and characters, the specific names, ranges, and behaviors of these types can vary between languages.



Sources:

Knudsen, J., & Niemeyer, P. (2005). Learning Java. O'Reilly.

Mughal, K. A., & Rasmussen, R. W. (2003). A programmer's guide to Java certification: a comprehensive primer. Addison-Wesley Professional.

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